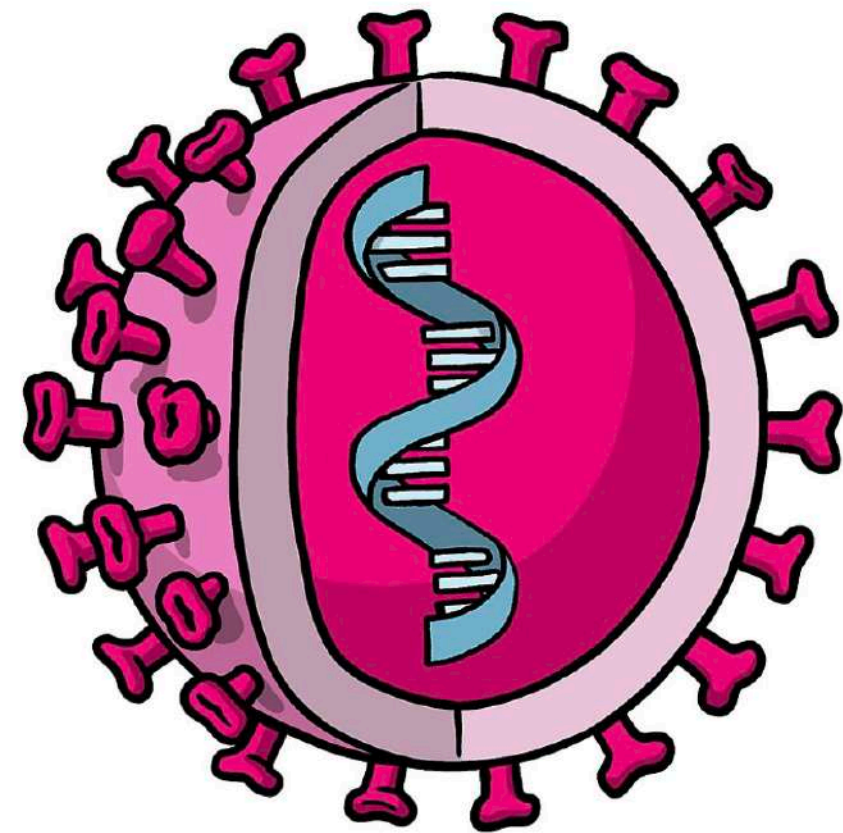
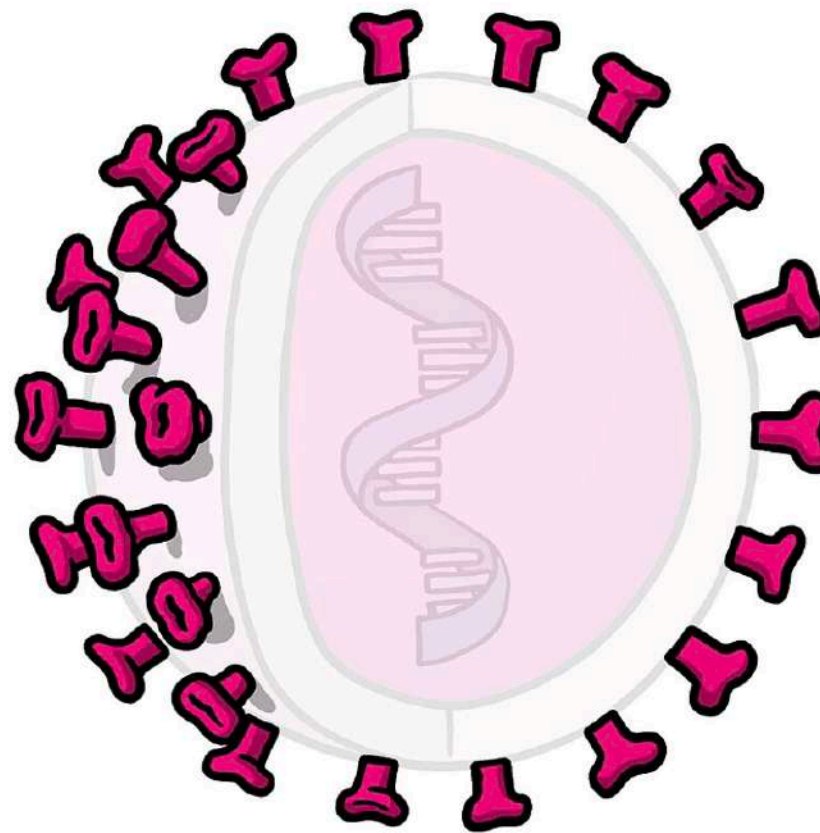




Using a whole virus
or bacterium

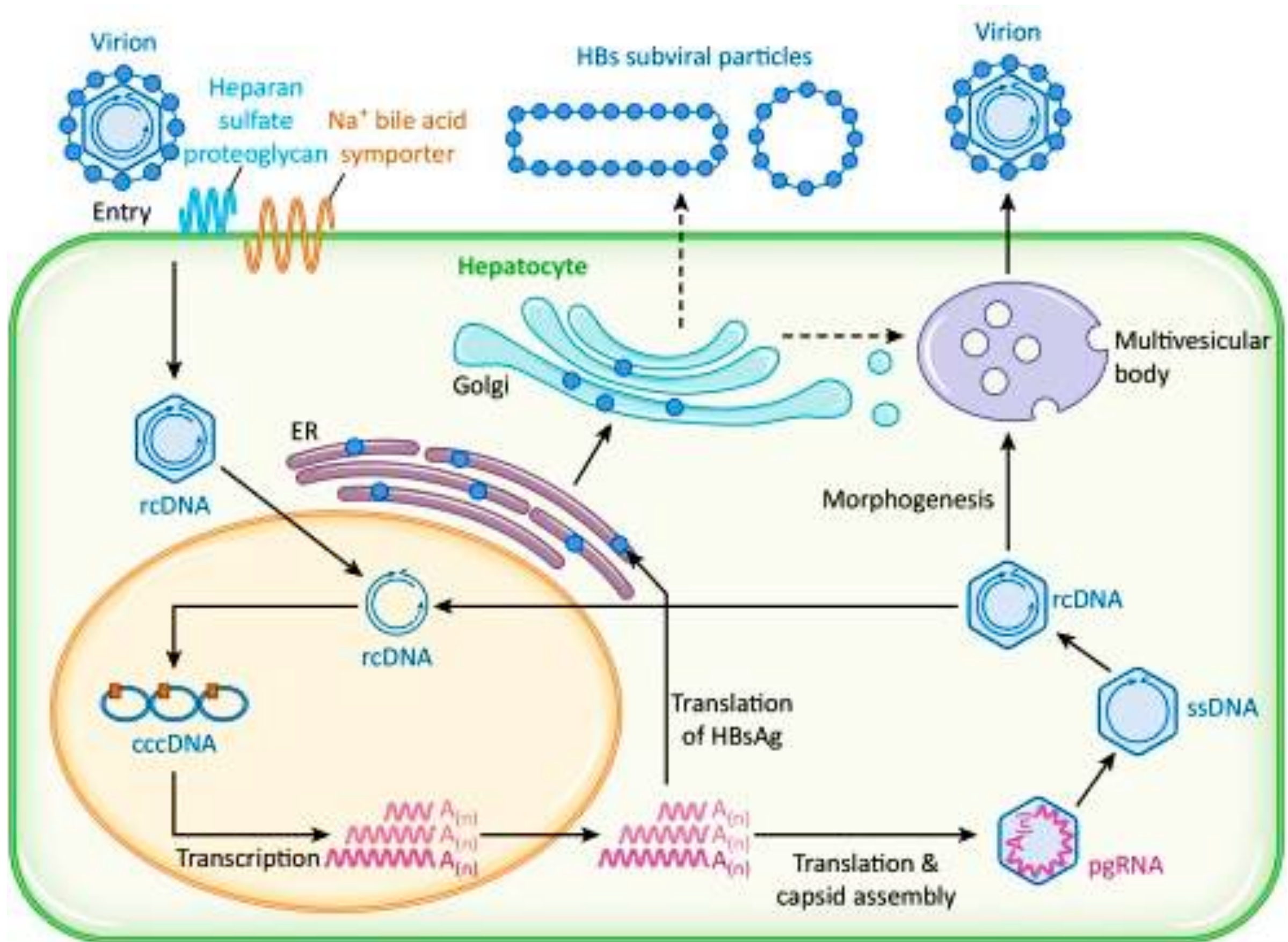


Parts that trigger
the immune system

- Needs to be attenuated
- Needs to re-learn what cells it likes to infect - in this process it may learn to infect cells which could cause other types of damage
- Must be grown in cells or tissues in large quantities or in whole embryos (chick)

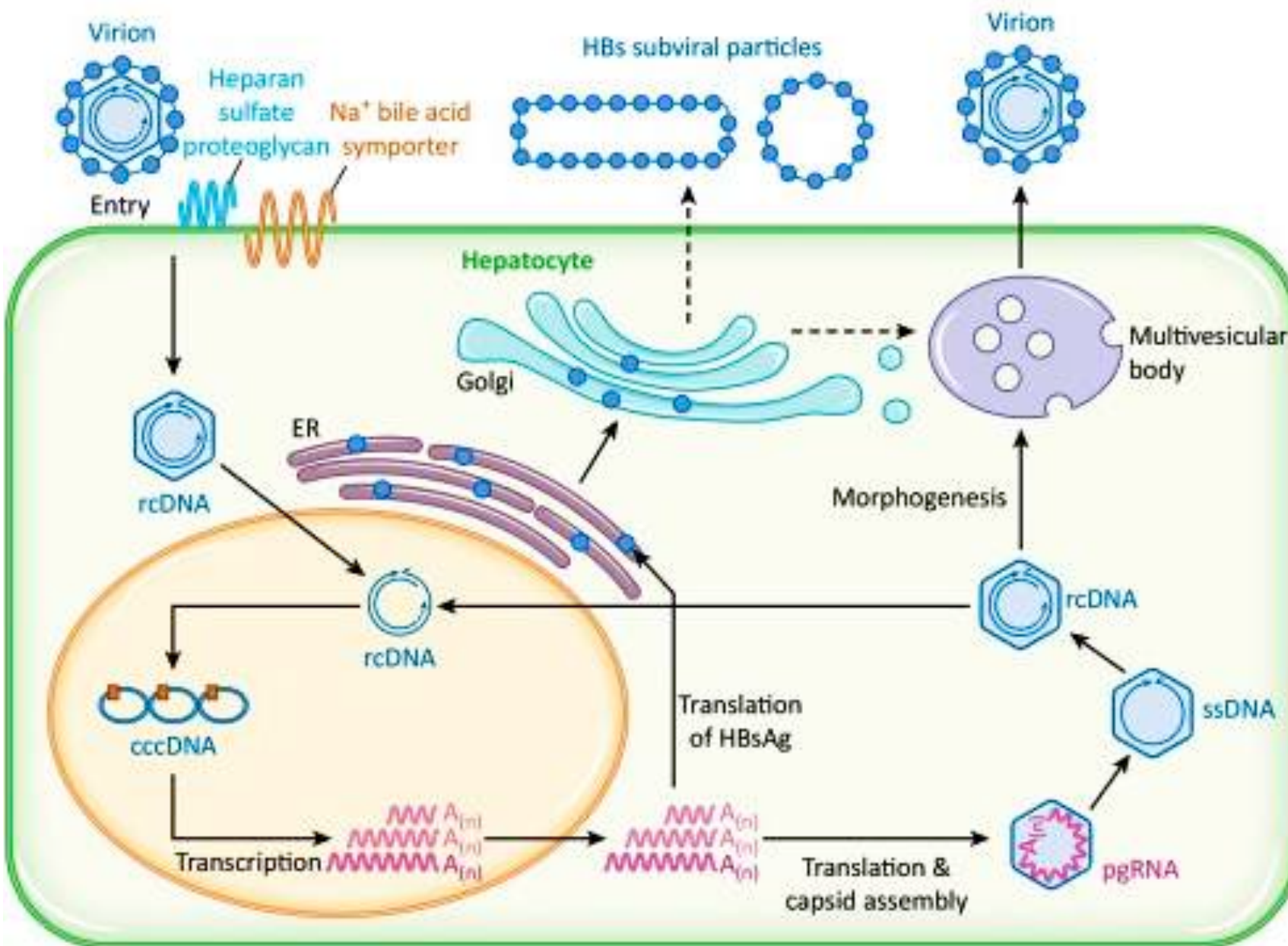
The basic idea is to trigger a “mild” infection to expose your immune system to the pathogen

Viruses - vaccination



Viruses - vaccination

Recombinant DNA technology for vaccine production



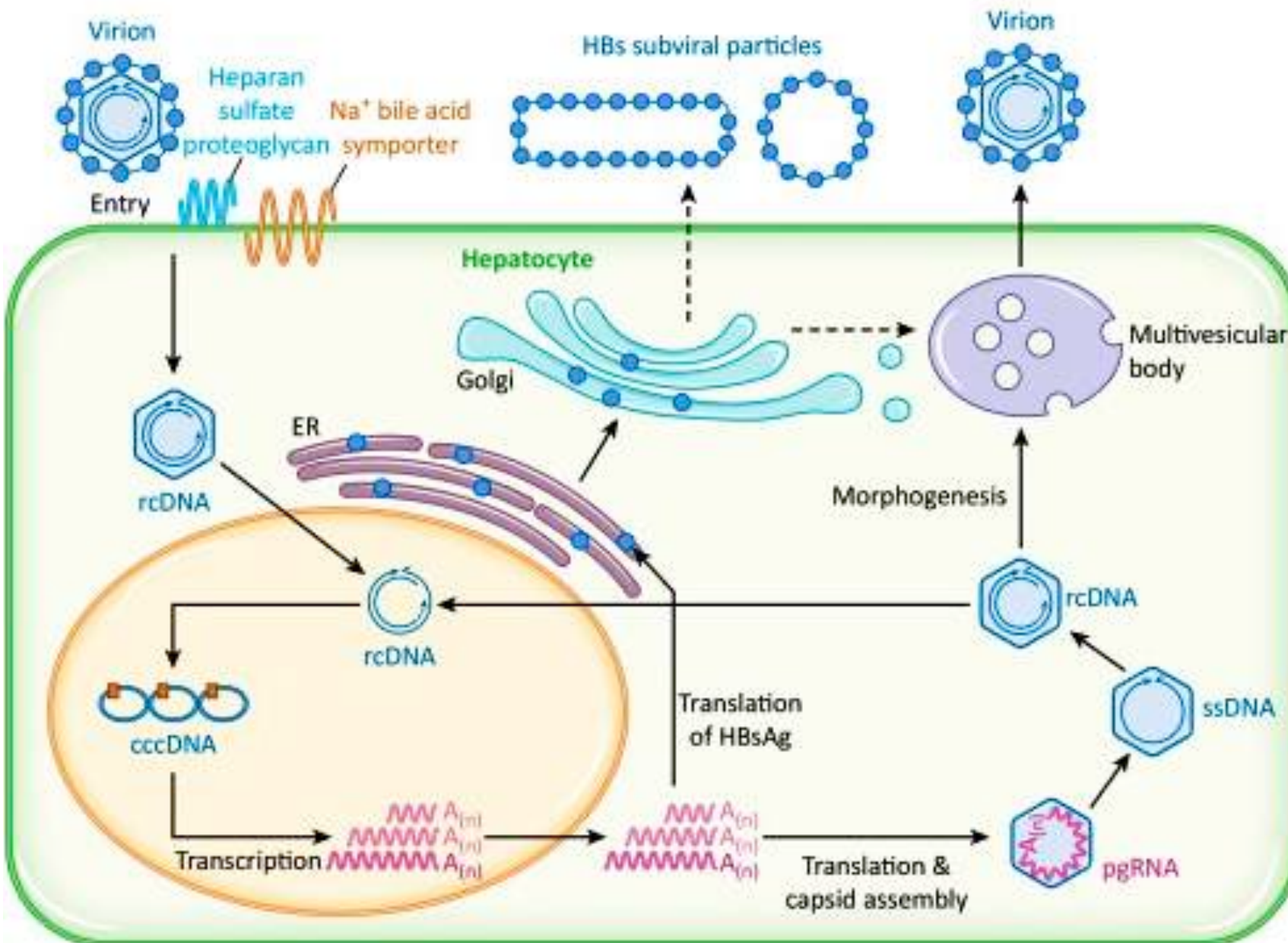
- HBV infection results in production of intact spherical virions ~42nm Dane particles, and **overproduction of 22nm particles of exclusively HBsAg**
- HBsAg is encoded by gene S and the large form of **HBsAg protein** is the most abundant form found on the surface of infectious viral particles crucial for HBV binding to hepatocytes
- **Cell-free production of immunogenic HBsAg in genome-free virus-like particles (VLPs)** paved the way for genetic engineering for human health

Empty shell of the virus, can trigger immune response in host, but cannot replicate and cause infection

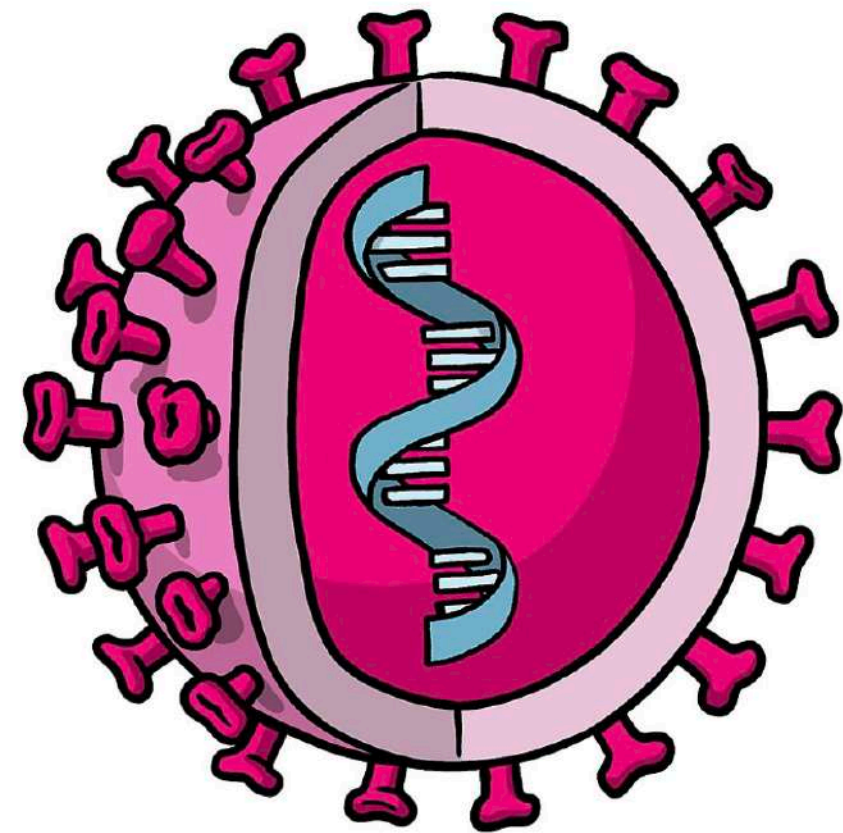
Viruses - vaccination

Recombinant DNA technology for vaccine production

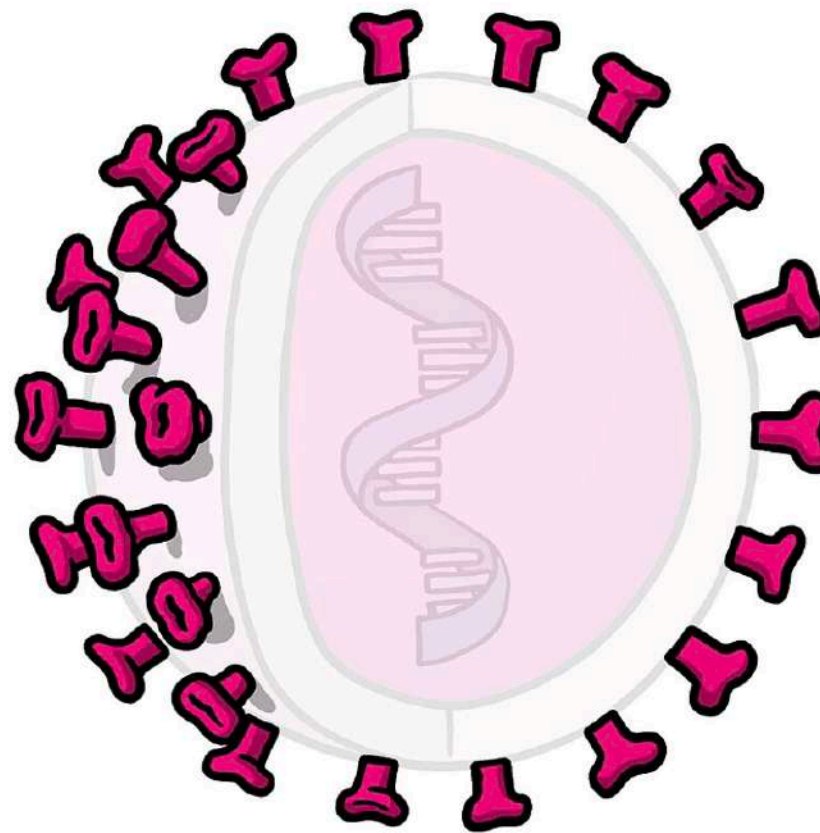
- preventative vaccine made of human Hepatitis B virus capsid protein (HBsAg)
- possible because the **gene for HBsAg was sequenced, cloned and protein produced in *E.coli* and Yeast**



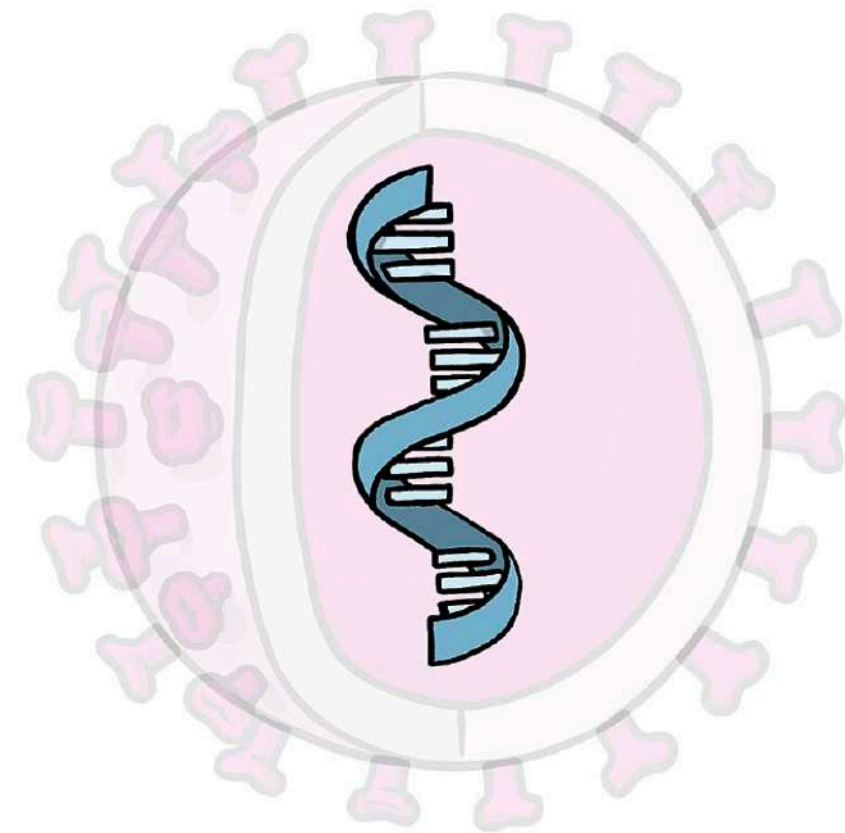
Empty shell of the virus, can trigger immune response in host, but cannot replicate and cause infection



Using a whole virus
or bacterium



Parts that trigger
the immune system



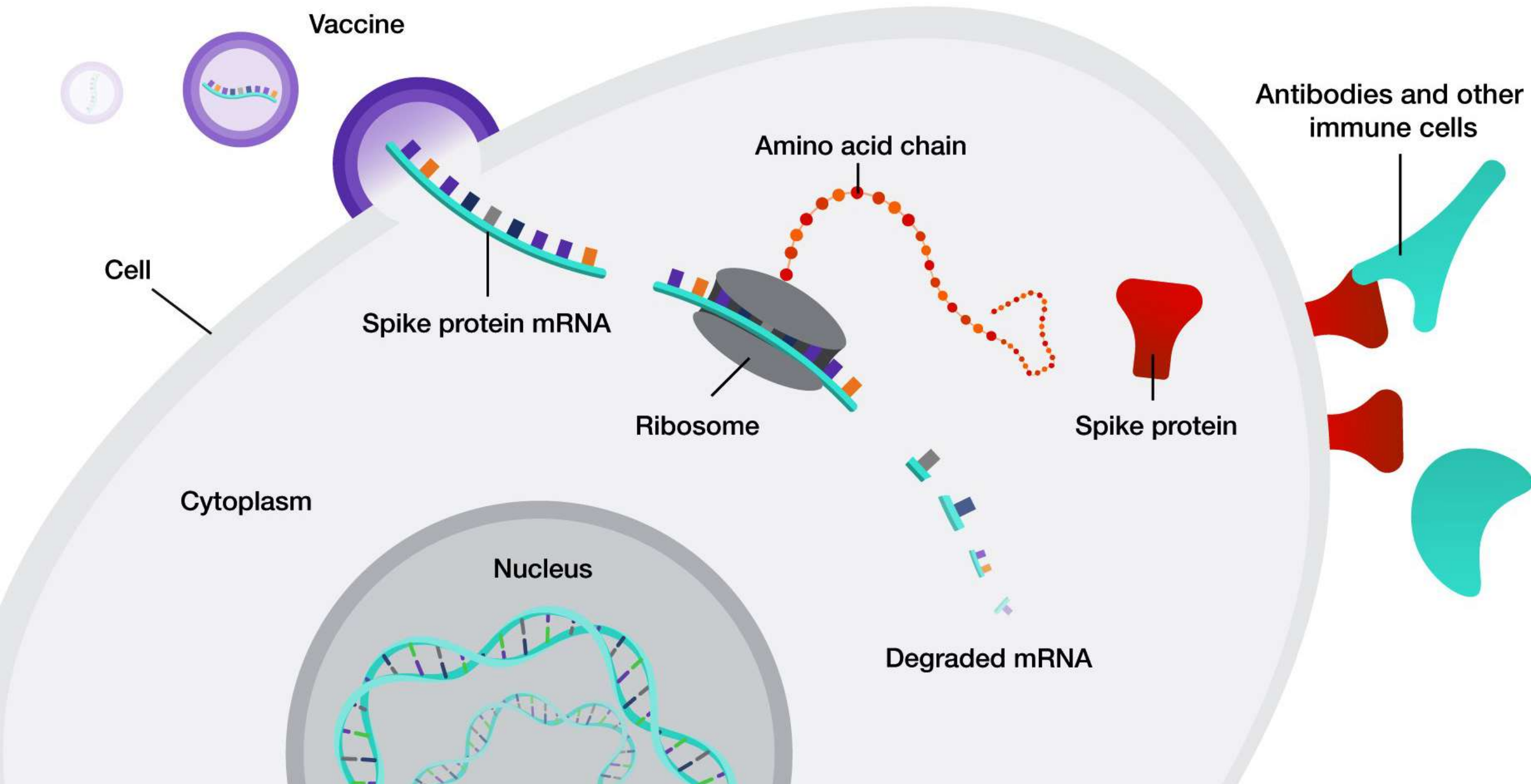
Just the
genetic material

- Needs to be attenuated
- Needs to re-learn what cells it likes to infect - in this process it may learn to infect cells which could cause other types of damage
- Must be grown in cells or tissues in large quantities or in whole embryos (chick)

The basic idea is to trigger a “mild” infection to expose your immune system to the pathogen

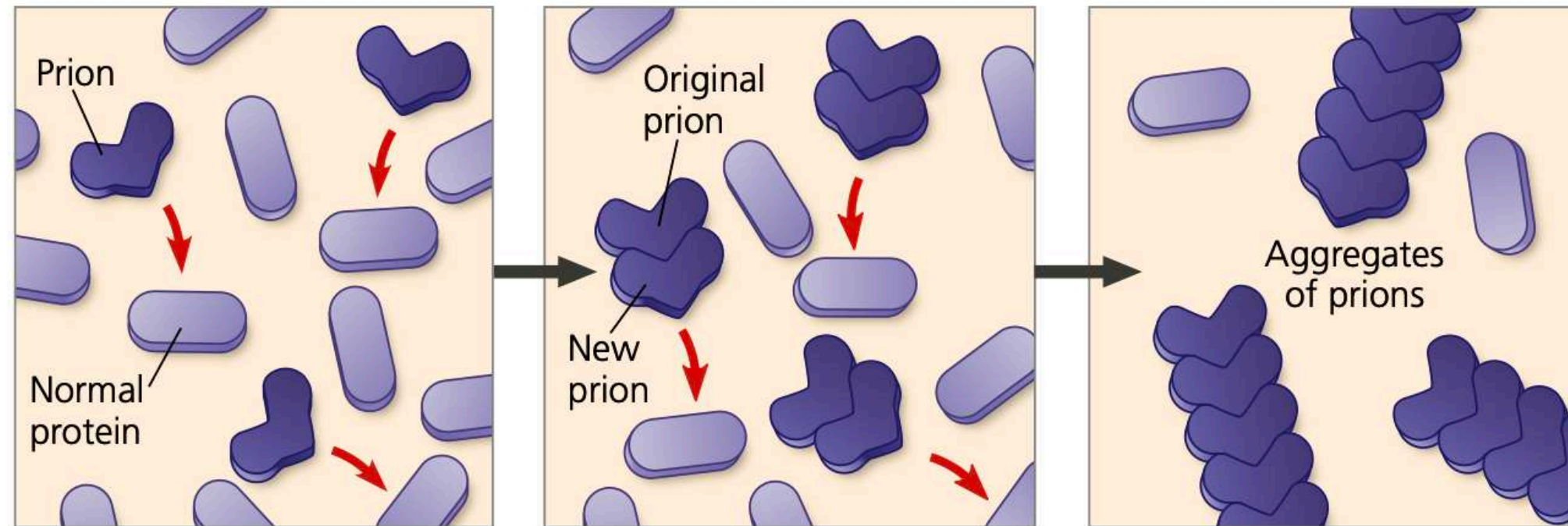
Concept of an mRNA vaccine

mRNA is packaged in liposomes or other nano-carriers



Infectious agents - only a biomolecule can also be infectious

- Viroids - circular RNAs of ~100 bases, infects plants, does not encode for proteins
- Prions - infectious proteins, cause degenerative brain diseases in animals



Prions are mis-folded versions of normal proteins. Somehow when they come in contact with their normal versions, they trigger a chain reaction, converting the normal version into a prion



Viruses infect all forms of life

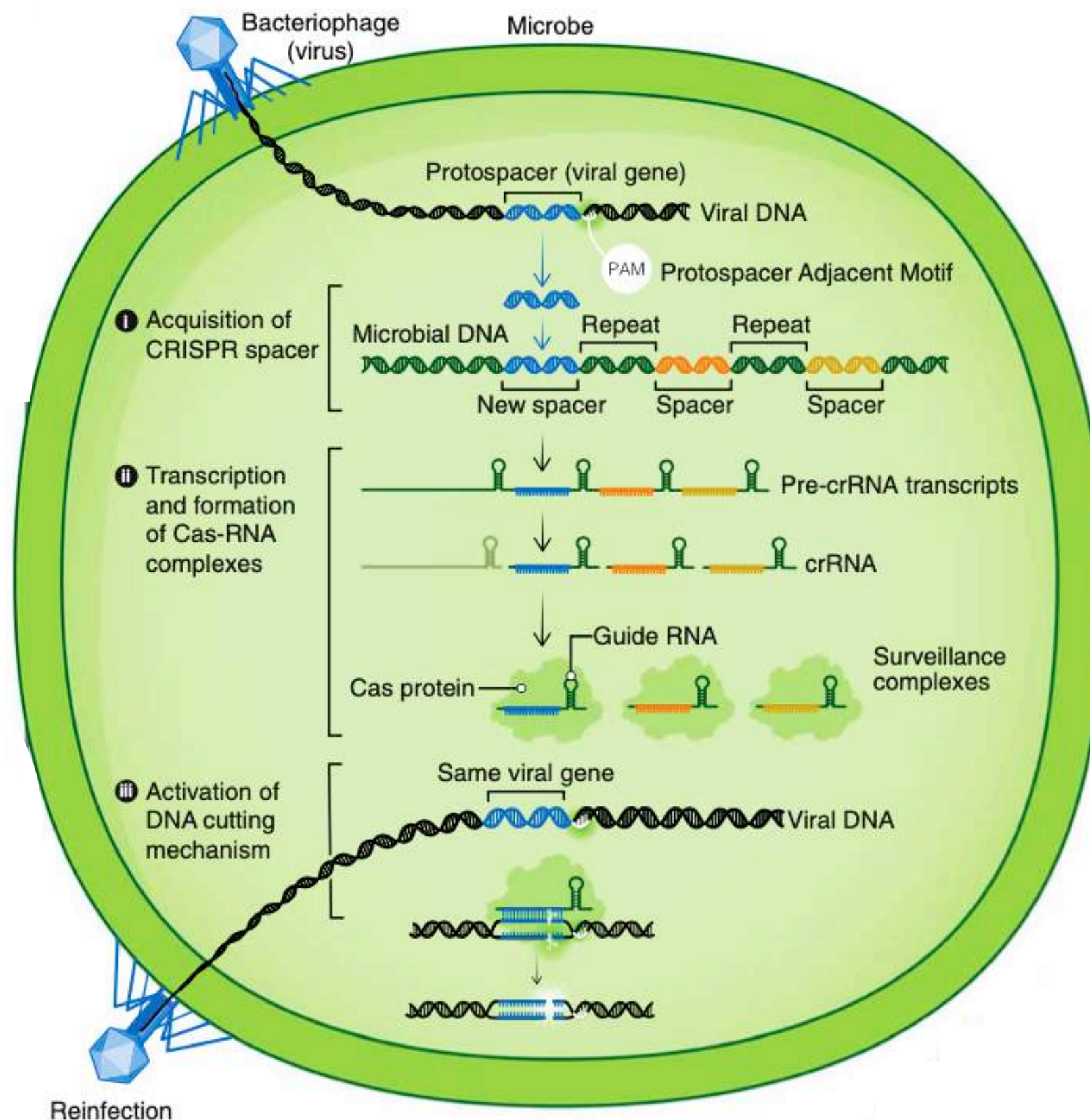
Organisms have evolved strategies to prevent or overcome infections caused by viruses

One such strategy is to trigger an immune response - requires a complex immune system

What about organisms that do not have an “immune system”?

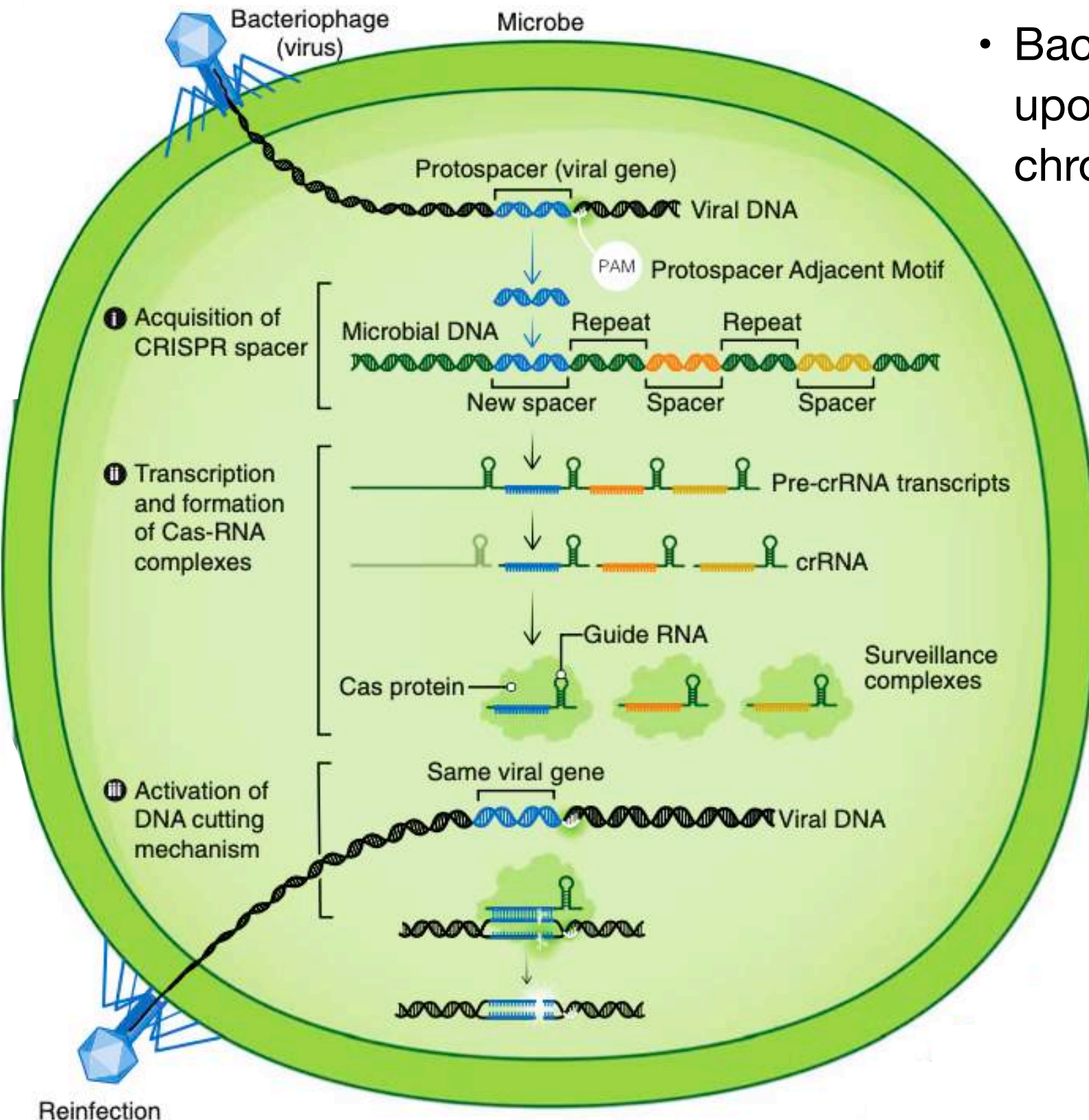
Viruses - how bacteria vaccinate themselves

Clusters of Regularly Interspaced Palindromic Repeats



Viruses - how bacteria vaccinate themselves

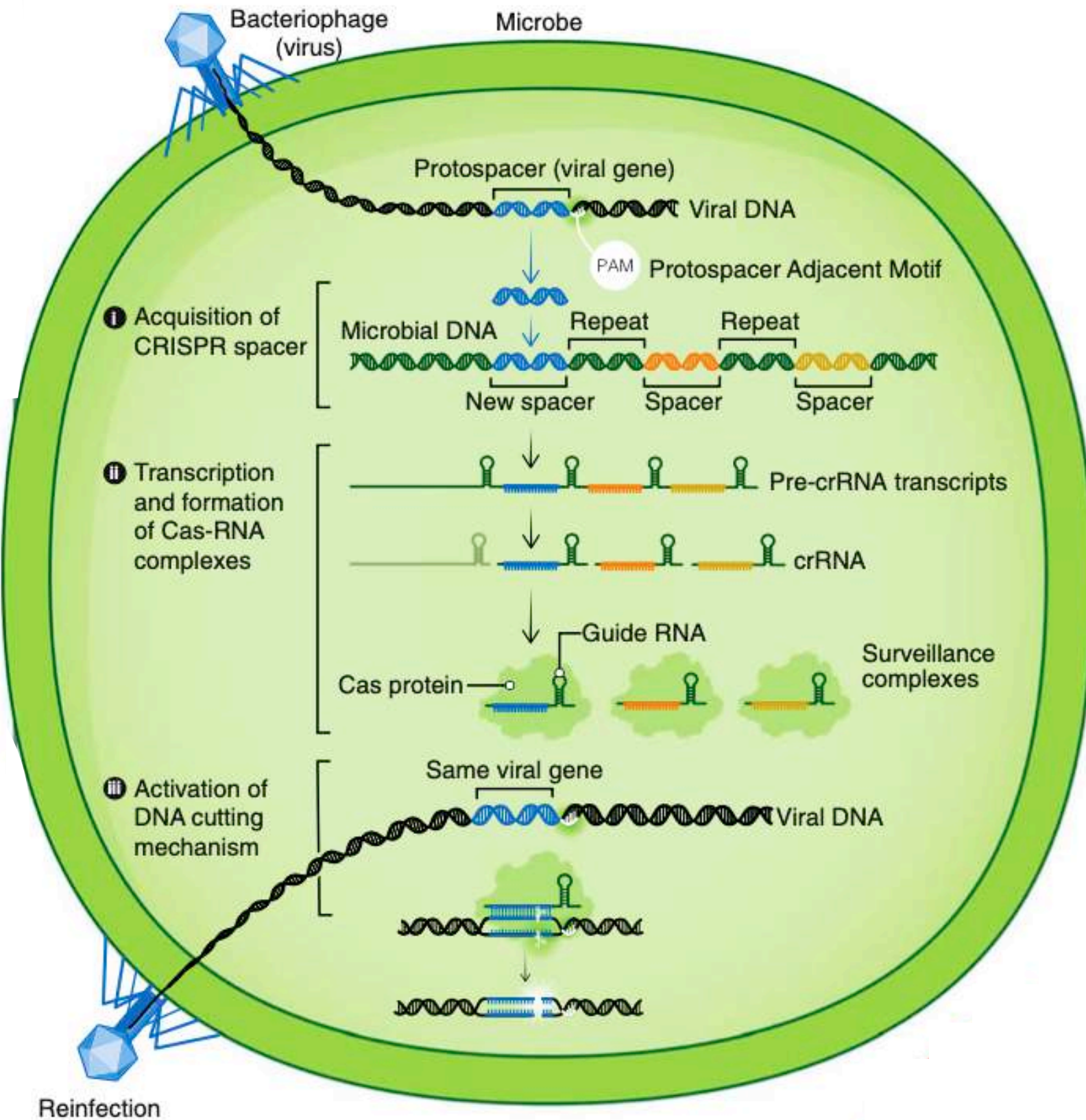
Clusters of Regularly Interspaced Palindromic Repeats



- Bacteria acquire viral genome sequences upon infection and insert it into their chromosome as spacer sequences
- Spacer sequences from different viruses are separated by bacterial DNA repeats
- Spacers are the “memory” of a bacteria’s encounter with specific viruses - acquired during an unsuccessful infection
- This “memory” enables the recognition and neutralization of the invading genetic material upon subsequent infections

Viruses - how bacteria vaccinate themselves

Clusters of Regularly Interspaced Palindromic Repeats



- Bacteria acquire viral genome sequences upon infection and insert it into their chromosome as spacer sequences
- CRISPR Associated genes (*cas* genes) convert the spacer+repeat into RNA
- Also makes *cas* protein, which is a nuclease (cuts DNA/RNA)
- The spacer+repeat RNA+*cas* protein complex “knows” where to target because of base complementarity between RNA sequence and DNA
- This complementarity recognizes the nucleic acid of the invading virus when encountered again

CRISPR - 2020 Nobel Prize in Chemistry

CRISPR discovery



Francisco Mojica
2005

CRISPR mode of action

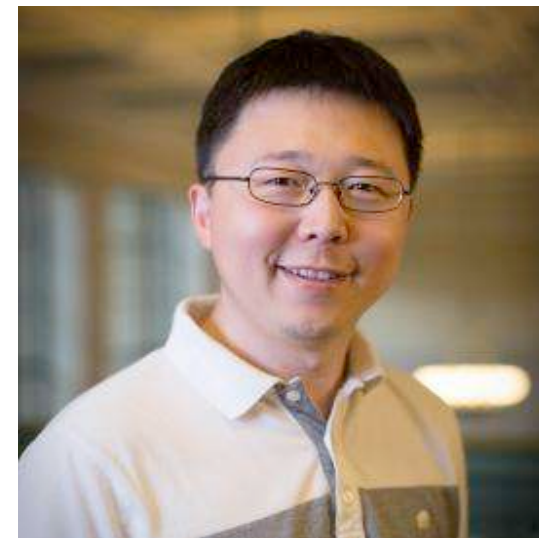


Emmanuelle Charpentier
2011

CRISPR application



Jennifer Doudna
2012



Feng Zhang
2013

Francisco Mojica originally identified the spacer DNAs from viral genomes in archaea and bacterial genomes when he was a PhD student

Mojica coined the term CRISPR

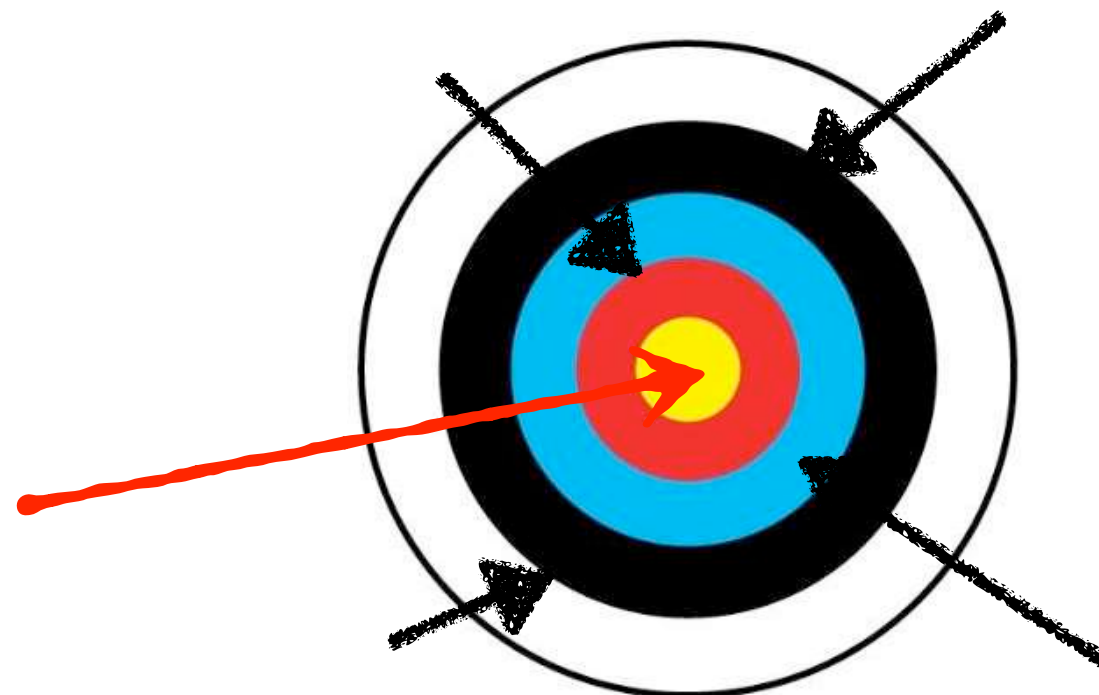
Why is CRISPR such a big deal?

- DNA can be “broken” in many ways
- Naturally occurring DNA mutations cause human diseases
- DNA mutations made in the lab in animals help mimic human diseases in model organisms
- Chemicals and radiations could “break” DNA and were engineered to be efficient

If DNA were a dart board, till 2012, we could ensure that the dart would hit somewhere on the board

The randomness of the hit meant that one could never target a specific site on the DNA

With CRISPR
we can hit
bulls eye



What makes CRISPR specific to a particular site?

If you know the sequence of DNA



You can predict the sequence of the RNA



You can make that RNA in vitro (guide RNA)



Mix the RNA with cas9 protein



The RNA-cas9 protein complex “knows where to go”

This occurs because of sequence complementarity between
RNA and DNA

Allows a particular RNA sequence to target a specific DNA sequence

Gene editing by CRISPR

- Genome editing by CRISPR-Cas requires a single guide (sg) RNA to direct the Cas9 endonuclease to a specific region of the genomic DNA
- Requires a PAM sequence (Protospacer Adjacent Motif), next to the site of cleavage
- Once it reaches the site on the DNA, Cas9 cuts the DNA resulting in a double strand break

The cuts caused by cas9 need to be repaired

